Coherent error: ZZ [bonus]

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Introduction to quantum noise

Coherent errors

Qiskit Global Summer School on Quantum Machine Learning

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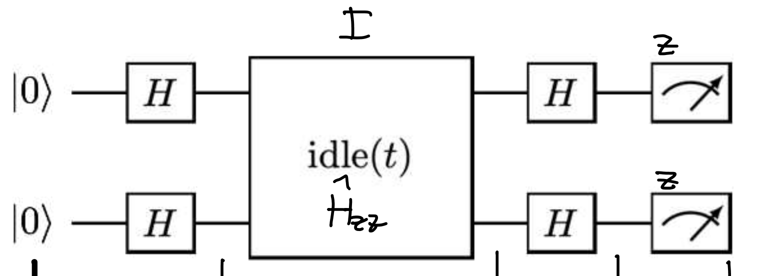

$|0\rangle|0\rangle$ $\quad$ $|X\rangle|X\rangle$, $|+\rangle|+\rangle$ $\quad$ $|X\rangle|X\rangle$ $\quad$ $|0\rangle|0\rangle$

Hadamard gate

$$H = \begin{array}{c|cc} & \langle 0| & \langle 1| \\ \hline |0\rangle & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ |1\rangle & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{array}$$

$$\begin{cases} H|0\rangle = |+x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \\ H|1\rangle = |-x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} \end{cases}$$

$$X|+X\rangle = +|+X\rangle$$
$$X|-X\rangle = -1|-X\rangle$$
$$|+X\rangle :=: |+\rangle$$
$$|-X\rangle :=: |-\rangle$$

$$\begin{cases} Z|0\rangle = +1|0\rangle \\ Z|1\rangle = -1|1\rangle \end{cases}$$

$$\langle ZI\rangle = +1$$
$$\langle IZ\rangle = +1$$
$$\langle ZZ\rangle = \langle 0|\langle 0| Z Z |0\rangle|0\rangle$$
$$= \langle 0|Z|0\rangle \langle 0|Z|0\rangle$$
$$= (+1)\ (+1)$$
$$= +1$$

Noisy

ZZ Interaction

$$\hat{H} = \frac{1}{2}\hbar\omega \hat{Z}\hat{Z}$$

$$\hat{U}(t) = \exp\left(-i\hbar^{-1}\hat{H}t\right)$$
$$= \exp\left(-i\frac{\omega t}{2}\hat{Z}\hat{Z}\right)$$
$$= \cos\left(\frac{\omega t}{2}\right)\hat{I} - i\sin\left(\frac{\omega t}{2}\right)\hat{Z}\hat{Z}$$
$$= \hat{R}_{ZZ}(\theta = \omega t)$$

$$\hat{R}_X(\theta) = \exp\left(-i\frac{\theta}{2}\hat{X}\right) \qquad X^2 = I$$
$$= \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)X$$

$$(\hat{Z}\hat{Z})^2 = Z^2 Z^2 = I_2 \otimes I_2 = I_4$$

$$|0\rangle|0\rangle \xrightarrow{HH} |+\rangle|+\rangle \xrightarrow{\text{idle}} R_{ZZ}(\theta)|+\rangle|+\rangle = \cos\frac{\theta}{2}|+\rangle|+\rangle - i\sin\frac{\theta}{2}|-\rangle|-\rangle \xrightarrow{HH} \underline{\cos\theta/2\,|0\rangle|0\rangle - i\sin\theta/2\,|1\rangle|1\rangle}$$

$$R_{ZZ}(\theta)|+\rangle|+\rangle = \cos\frac{\theta}{2}|+\rangle|+\rangle - i\sin\frac{\theta}{2}(Z|+\rangle)_1 \otimes (Z|+\rangle)_2$$

$$Z|+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} = |-\rangle$$

$$Z|+\rangle = |-\rangle$$
$$Z|-\rangle = |+\rangle$$

$$\langle ZI\rangle = \cos\omega t$$
$$\langle IZ\rangle = \cos\omega t$$
$$\langle ZZ\rangle = 1$$